

SegKAN: High-Resolution Medical Image Segmentation with Long-Distance Dependencies

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Abstract—Hepatic vessels in computed tomography scans often suffer from image fragmentation and noise interference, making it difficult to maintain vessel integrity and posing significant challenges for vessel segmentation. To address this issue, we propose an innovative model: SegKAN. First, we improve the conventional embedding module by adopting a novel convolutional network structure for image embedding, which smooths out image noise and prevents issues such as gradient explosion in subsequent stages. Next, we transform the spatial relationships between Patch blocks into temporal relationships to solve the problem of capturing positional relationships between Patch blocks in traditional Vision Transformer models. We conducted experiments on a Hepatic vessel dataset, and compared to the existing state-of-the-art model, the Dice score improved by 1.78%. These results demonstrate that the proposed new structure effectively enhances the segmentation performance of high-resolution extended objects. Code will be available at <https://github.com/goblin327/SegKAN>.

Index Terms—Hepatic vessels, Global context modeling, temporal sequence network, elongated structure segmentation

I. INTRODUCTION

Medical image segmentation [1] plays a crucial role in extracting important information from computed tomography (CT) images, providing significant support for disease diagnosis, radiation therapy localization, and surgical planning [2]. Therefore, in recent years, various computational models have been proposed to tackle this task, including traditional machine learning methods and advanced deep learning techniques [3]–[5]. However, abdominal medical images typically have high resolution, high dimensionality, complex tissue structures, and multimodal characteristics, while also suffering from noise, artifacts, and challenges related to standardization, all of which pose significant challenges for segmentation tasks [6]–[8]. Especially in vessel structure segmentation, a common issue is the inability to achieve long-distance continuous segmentation

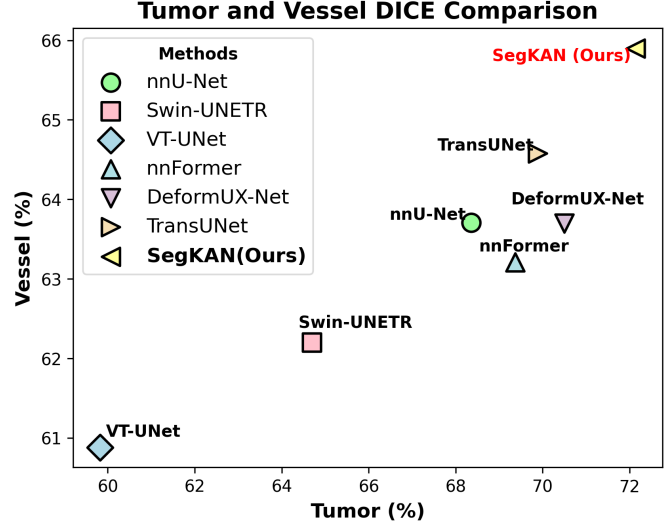


Fig. 1. Comparison of Tumor and Vessel Segmentation Results Across Different Models.

[9]. For example, in Figure 2, the blood vessels in the original image typically exhibit elongated and continuous structures. However, due to breakages, noise interference, or resolution limitations, the integrity of the blood vessels is difficult to maintain, which presents a serious challenge for learning vessel features. Furthermore, as shown in Figure 2, many blood vessels are intricate, interwoven, and densely distributed, resulting in very limited information that can be extracted. Additionally, because individual blood vessels occupy small areas, the task of delineating blood vessels is particularly cumbersome and lacks large-scale, high-quality annotated data. These challenges collectively lead to suboptimal performance of existing models in vessel segmentation tasks.

To address this issue, traditional image processing methods such as thresholding, region growing, and edge detection mainly rely on features of vessel pixels and boundaries [10]. However, despite the dozens of methods proposed from various perspectives, the performance of Hepatic vessel segmentation remains short of expectations. In fact, hepatic vessel segmentation has always been a challenging task, especially since blood vessels often exhibit high resolution and low contrast, leading to blurred boundaries, loss of detail, and confusion with surrounding tissues, making the segmentation process extremely difficult [11], [12]. When dealing with

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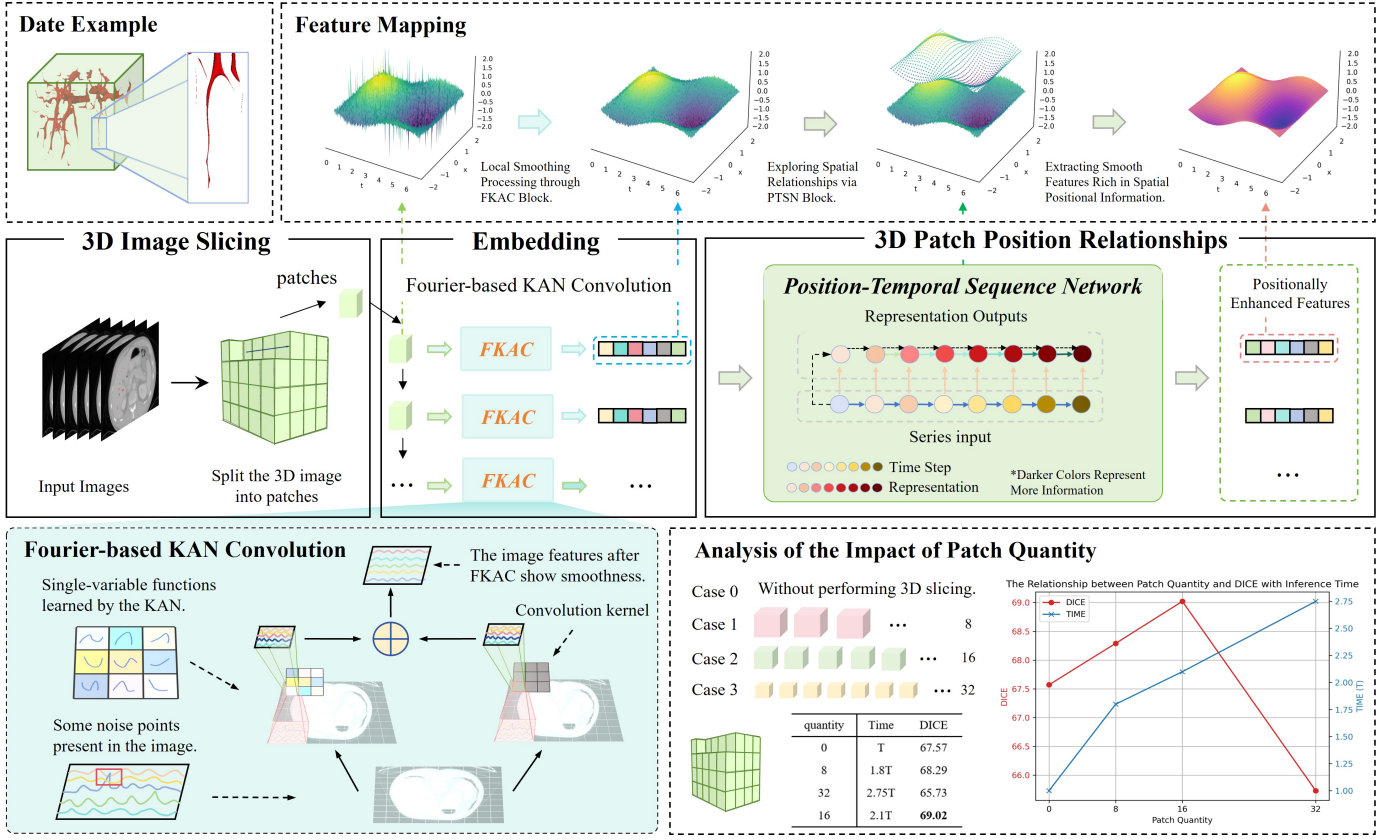


Fig. 2. The core algorithm framework of SegKAN, illustrates the 3D image slices and encoding of 3D patches, as well as the learning of spatial relationships between 3D patches. The unique elongated structure of Hepatic blood vessels is highlighted in the top-left corner. The bottom-right image shows the impact of different split quantities on the model's segmentation results. The model performs best when the split quantity reaches 16, indicating that an appropriate number of splits can significantly improve segmentation performance in the SegKAN model.

complex vessel structures and high-resolution image data, these traditional methods exhibit significant limitations.

With the rapid development of AI for health [13]–[17], deep learning-based image detection [18]–[20] and segmentation methods [21]–[24] have made significant progress in Hepatic vessel segmentation. In recent years, researchers have introduced techniques such as self-attention mechanisms and cross-scale feature fusion to significantly enhance the model's ability to handle long-distance connections and complex structures [25]–[29]. These methods can effectively capture global information about blood vessels and strengthen the associations between different regions, improving the segmentation of long-distance blood vessels. However, current segmentation methods still struggle to effectively extract long-distance vessel structures in high-resolution images [30], [31]. Therefore, there is an urgent need for an innovative approach that can effectively address the long-distance segmentation problem in high-resolution images.

To solve the aforementioned problem, this paper proposes a novel medical image segmentation model: SegKAN. As shown in Figure 2, the model first divides 3D images into multiple 3D patches and uses a time-series network to transform the spatial relationships between these patches into sequential relationships, thus solving the problem in traditional Vision

Transformer (ViT) models where the positional relationships between patches are difficult to capture. We innovatively combine the spatial relationships in the image with time-series data and design a brand-new network structure to tackle the challenges in long organ segmentation tasks. This approach not only effectively reduces the loss of small-scale feature information in medical image segmentation but also better models global context, thus improving segmentation accuracy.

In summary, the contributions of this paper are as follows:

- We propose PTSN (Position-Temporal Sequence Network), which effectively optimizes the relative positional relationships between patches in an image. Specifically, we transform traditional spatial positional relationships into temporal sequence relationships. By inputting the patch sequence into the model in a temporal sequence manner, we further reinforce the positional relationship between patches, thereby improving the model's performance in long-distance segmentation tasks.
- We introduce FKAC (Fourier-based KAN Convolution), an innovation built upon traditional embedding methods. This method combines the superior fitting ability of the KAN network with convolutional operations, successfully smoothing the noise in local regions and preventing issues such as gradient explosion during network training. This

approach effectively enhances the model's stability and generalization ability.

- For the elongated structures of hepatic vessels, we validated our method on the Hepatic Vessel dataset and achieved results surpassing the current state-of-the-art method, TransUNet. Experimental results demonstrate that, within Transformer-based architectures, our method further optimizes contextual modeling ability, thereby improving long-distance segmentation performance.

II. RELATED WORKS

The blood vessels in the Hepatic appear as tiny spots in cross-section, due to the vessel branches being elongated tubular structures. Therefore, the task of localizing blood vessels in the Hepatic is similar to the segmentation problem of small objects. However, segmentation of small objects faces many challenges, such as information loss, noise interference, insufficient samples, and high sensitivity to disturbances [32]. Traditional transformer-based methods either fail to effectively capture semantic information [33], [34] due to the point-to-point attention mechanism [35], [36] or suffer from high computational complexity, which limits their performance. To address this issue, recent studies have introduced effective solutions by incorporating image patches. Based on this, SwinUNet and SwinUNETR optimize the self-attention mechanism using the more computationally efficient Swin Transformers; nnFormer [37] further improves model performance by alternating convolution and self-attention. Additionally, Meng et al. [38] combined multi-scale feature fusion with a transformer and proposed a method for fine segmentation of Hepatic tumors and blood vessels. These studies achieve a good balance between accuracy and computational efficiency.

In recent years, Liu et al. [39] proposed the Kolmogorov-Arnold Networks (KAN), which have made significant research progress. This novel neural network architecture has gained widespread attention due to its advantages in interpretability and accuracy [40]. However, due to issues such as a large number of parameters and slower inference speed, Dong et al. [41] proposed the FAN network as an improvement. FAN combines the Fourier series to model periodic phenomena and demonstrates superior performance in various practical tasks. At the same time, Yang et al. [42] introduced the KAT model, which replaces the traditional MLP layer with KAN layers, addressing a series of challenges including parallel computation, parameter efficiency, and weight initialization, thus enhancing the expressive power and performance of the model. Given the strong fitting ability of the KAN network, Pal et al. [43] further proposed MULTKAN, while Bozorgasl et al. [44] introduced Wav-KAN. Both models were successfully applied to the ODE/PDE identification tasks of nonlinear dynamic systems by combining the SRDD denoising algorithm and the PISF algorithm, respectively. Additionally, Wav-KAN enhanced the network's multi-resolution analysis capability by incorporating wavelet transforms. Finally, Li et al. [45] proposed the XNET model based on KAN, inspired by the Cauchy Integral Theorem. The model uses the Cauchy kernel as a basis function and

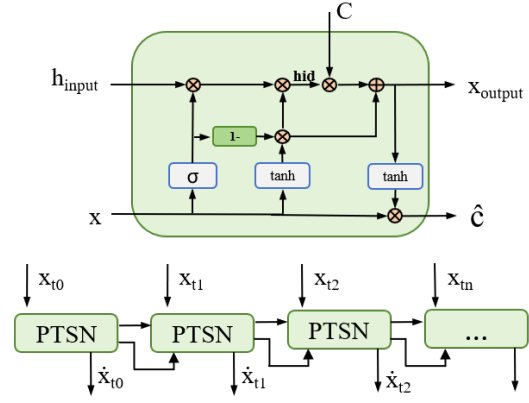


Fig. 3. PTSN Structure

replaces the feedforward neural network (FNN) component in LSTM with XNet, successfully capturing complex data features. This model surpasses traditional MLP and KAN in both speed and accuracy, demonstrating its great potential.

III. METHODS

A. paradigm—Position-Time Series Network (PTSN)

The core advantage of Recurrent Neural Networks (RNNs) lies in their recursive structure, which uses the output of previous time steps as input for the current step, thereby gradually accumulating contextual information from different locations. When processing images, RNNs can effectively capture the spatial relationships between patches in the image and model long-range dependencies.

Building on this advantage, we propose a novel general paradigm—Position-Time Series Network (PTSN). As shown in Figure 3, this design employs a gated mechanism combined with the σ (sigmoid) and $\tanh$ activation functions to dynamically filter and suppress feature information. At the same time, by leveraging the complementary relationship between the "1—" operation and the sigmoid output, the network dynamically adjusts the flow of features, dividing them into key parts to be preserved and ineffective parts to be suppressed, thus enabling more refined feature selection. This design helps balance information flow, preventing features from being overly concentrated or completely lost, while enhancing the model's nonlinear expressiveness by separating the retained and suppressed information. Additionally, the mechanism reduces noise interference, lowers the risk of overfitting, and improves the model's stability and adaptability to complex data through the diversion operation during gradient backpropagation. The computation formula is:

$$G = \sigma(x) \quad (1)$$

$$H = \tanh(x) \quad (2)$$

$$hid = h_{input} \odot G + (1 - G) \odot H \quad (3)$$

$$c = hid \odot c_{input} + G \odot H \quad (4)$$

$$h = hid \odot \tanh(c) \quad (5)$$

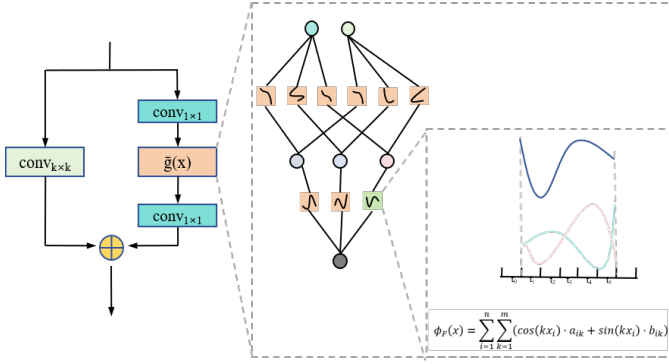


Fig. 4. FKAC Structure

B. Fourier-based KAN Convolution (FKAC)

Based on previous analysis, the limitations of PTSN are primarily attributed to the recursive structure of RNNs, especially the issue of gradient explosion during the backpropagation process. In contrast, convolution operations can effectively smooth input data, filter out high-frequency noise, and remove redundant details, thus mitigating this problem. This characteristic helps reduce the burden on the PTSN layer, allowing it to focus more on learning the key features of the sequence. By applying convolution operations to the input data, extreme values in the data can be smoothed or weakened, thereby reducing the magnitude of gradient fluctuations during backpropagation. As a result, we introduced an optimization in the embedding operation of VIT by adopting the KAN convolution method.

The main difference between KAN convolution and the conventional convolution used in traditional CNNs lies in the construction of the convolution kernel. In traditional CNNs, the convolution kernel is composed of weight parameters, whereas in KAN convolution, each element φ of the convolution kernel is a learnable nonlinear function represented using B-splines. Formally, each φ is a univariate nonlinear function with trainable parameters. In the original paper, the authors proposed the following form for the function:

$$\varphi = w_b \cdot \text{SiLU}(x) = \frac{x}{1 + e^{-x}} + w_s \cdot \text{Spline}(x) \quad (6)$$

In KAN convolution, the convolution kernel slides over the image, applying the corresponding activation function φ_{ij} to the respective pixels a , and the output pixel is computed as the sum of $\varphi_{ij}(a_{kl})$. Let K be a KAN convolution kernel, where $K \in \mathbb{R}^{N \times M}$, and the image is represented as a matrix:

$$(\text{Image} * K)_{i,j} = \sum_{k=1}^N \sum_{l=1}^M \phi_{kl}(a_{i+k,j+l}) \quad (7)$$

Although KAN convolution performs better in terms of fitting accuracy, its high computational complexity and large number of parameters cannot be overlooked. The main issue with KAN convolution lies in the basis functions within

the model, which require more computational resources during training and increase the risk of overfitting. To address these issues, we propose the Fourier-based KAN Convolution (FKAC), as shown in Figure 4.

Before applying the basis functions to the input data, we suggest first using a compression convolution with a kernel size of 1, followed by the application of the basis functions, and then an expansion convolution with a kernel size of 1. Intuitively, this process can be seen as a single-layer encoder, which first extracts meaningful features from the input data, then processes them through the chosen basis functions, and finally restores them via a single-layer decoder. The residual activation helps preserve important details that may be lost during the encoding and decoding process.

However, due to the introduction of spline functions, training KAN is more difficult than training an MLP, which contradicts the more efficient and effective alternative we propose. The essence of KAN is to approximate any function through the sum of multiple nonlinear functions. Our goal is to decompose a complex function into several relatively simple nonlinear functions. Naturally, Fourier coefficients become an attractive choice. Therefore, we propose the following equation:

$$\phi_F(\mathbf{x}) = \sum_{i=1}^d \sum_{k=1}^g (\cos(k\mathbf{x}_i) \cdot a_{ik} + \sin(k\mathbf{x}_i) \cdot b_{ik}) \quad (8)$$

IV. EXPERIMENTS

A. Dataset

The Hepatic Vessel dataset is sourced from the Medical Segmentation Decathlon (MSD) challenge, a competition focused on medical image segmentation. Its primary goal is to segment hepatic vessels and tumors from Hepatic CT images. The dataset contains 443 cases of 3D CT data, with each slice image having a resolution of 512x512 pixels. Due to the elongated and intricate structure of hepatic vessels, the HepaticVessel dataset is particularly suitable for analyzing the performance of models in long-distance segmentation tasks.

B. Experiment Setup

All experiments were conducted on a device equipped with an NVIDIA 3090 GPU (24GB memory) using the PyTorch framework. For the MSD dataset, five-fold cross-validation was employed for evaluation. The initial learning rate was set to 0.01, the batch size was 2, and the model was trained for 1000 iterations. For performance evaluation, the Dice similarity coefficient was used as the primary metric to measure the overlap between the predicted results and the ground truth labels.

C. Experimental Results

The experimental results on the Hepatic Vessel dataset indicate that, due to the large volume of the dataset, models based on self-attention mechanisms significantly outperform pure convolutional models, such as nnUNet, especially in

the recent nnFormer and TransUNet models. Thanks to the introduction of the self-attention mechanism, they achieved excellent scores of 66.29% and 67.24%, respectively. This demonstrates that the self-attention mechanism can effectively model global context on high-resolution images. However, these models overlook the learning of relative positions between patches, failing to fully leverage the advantages of long-range dependency modeling. To further improve segmentation performance, our method enhances the modeling of positional information between patches based on the self-attention mechanism, achieving an average DICE score of 69.02%. Particularly in the segmentation accuracy of hepatic vessels, our model achieved 65.90%, an improvement of 1.32% over TransUNet, as shown in Table I and Figure 1.

TABLE I
MODEL PERFORMANCE COMPARISON. THE BEST SCORES ARE INDICATED IN **BOLD**.

Method	Vessel	Tumor	Average
nnU-Net [46]	63.71	68.36	66.04
Swin-UNETR [47]	62.20	64.70	63.50
VT-UNet [48]	60.88	59.82	60.35
nnFormer [49]	63.21	69.37	66.29
DeformUX-Net [50]	63.70	<u>70.50</u>	67.10
TransUNet [51]	<u>64.58</u>	69.89	<u>67.24</u>
SegKAN (Ours)	65.90	72.14	69.02

D. Ablation Study

The proposed SegKAN model is primarily composed of two key modules: Fourier-based KAN Convolution (FKAC) and Paradigm—Position-Time Series Network (PTSN). The FKAC module aims to optimize the learning of noisy regions, preventing issues such as gradient explosion in subsequent models, thus improving the model’s stability. The PTSN module, on the other hand, converts the relative positional relationships between patches into temporal relationships, further enhancing the model’s ability to model relative positions. To study the effects of these two modules, we switched them individually while keeping the other parts unchanged. Experimental results show that the combination of FKAC and PTSN yields the best performance, with the model’s DSC (Dice Similarity Coefficient) improving by 2.54% compared to the baseline. More details can be found in Table II.

TABLE II
ABLATION STUDY OF SEGKAN. THE BEST SCORES ARE INDICATED IN **BOLD**.

Method	FKAC	PTSN	Vessel	Tumor	Average
SegKAN	✓	×	<u>63.70</u>	71.43	<u>67.57</u>
	×	✓	62.20	71.94	67.07
	×	×	62.51	70.63	66.57
	✓	✓	65.90	72.14	69.02

In this experiment, the temporal network processes the patch positional information by encoding it as a time series. We conducted several experiments to compare the impact of

various temporal networks on long-distance segmentation performance. The experimental results show that PTSN performs the best among all methods, significantly outperforming other network structures, as shown in Table III.

TABLE III
COMPARISON OF DIFFERENT TEMPORAL METHODS IN SEGKAN. THE BEST SCORES ARE INDICATED IN **BOLD**.

Method	Method	Vessel	Tumor	Average
SegKAN	LSTM [52]	<u>65.76</u>	<u>71.34</u>	<u>68.55</u>
	SimpTime [53]	64.01	69.64	66.83
	mamba [54]	64.82	69.38	67.10
	×	62.51	70.63	66.57
	PTSN (Ours)	65.90	72.14	69.02

To analyze the sensitivity of PTSN to the number of 3D image segments, we tested the performance of PTSN under different patch quantities, as shown in Table IV and Figure 2. The experimental results indicate that as the number of patches increases, the inference time also increases. However, increasing the number of patches allows for better segmentation of small targets, and with the introduction of PTSN, the average DICE score improves. When the number of patches is 16, the average DICE reaches 69.02%. However, when the number of patches increases to 32, due to the size limitations of the image, the number of segmented patches becomes too large, and the information in each patch becomes too limited, leading to a decrease in the average DICE to 65.73%. This result further indicates that the number of patches in PTSN has a certain impact on the final performance, and an appropriate number of patches has a positive effect on model performance.

TABLE IV
ABLATION EXPERIMENT WITH DIFFERENT PATCH QUANTITIES. THE BEST SCORES ARE INDICATED IN **BOLD**.

Method	Patch Quantity	Vessel	Tumor	Time	Average
PTSN	8	63.68	72.90	1.8T	68.29
	32	60.72	70.74	2.75T	65.73
	×	<u>63.70</u>	71.43	T	67.57
	16 (Ours)	65.90	<u>72.14</u>	2.1T	69.02

V. CONCLUSIONS

This paper introduces the SegKAN model, which not only effectively reduces local noise and enhances the stability of the model but also optimizes the long-distance segmentation of thin structures (such as blood vessels) in medical images. By incorporating the PTSN module, the model transforms the spatial relationships between patch blocks into temporal relationships and enhances the relative positional relationships between patch blocks through time-series modeling. To prevent issues such as gradient explosion, we propose the FKAC module, which optimizes the traditional embedding module. Based on this, we combine the KAN network, known for its excellent fitting capabilities, and replace the traditional embedding layer with residual convolution, successfully achieving

long-distance segmentation in complex environments. Extensive experimental results on the Hepatic Vessel dataset show that the SegKAN model outperforms existing state-of-the-art methods. Future research will further explore the potential applications of SegKAN in more complex and diverse medical image environments.

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